G Future Work

The current version of SpisNemt proves itself feasible of native machine learning for food waste reduction, there is still room for technical and functional expansion. The following contents outline a possible roadmap for transitioning the system from a prototype, to a production-ready application.

G.1 Inception V3 flexibility

The generalization gap identified in section 14 needs a specialized training method. Future work in this regard involves a strategy, where the Inception V3 model is tuned on a dataset captured specifically in kitchen environments rather than grocery store and professional photo studio settings. This would address the probable issues of background noise and bad lighting that impact the classification accuracy.

Additionally, expanding the model's image dataset to include a broader ingredient database eventually moving beyond vegetables to include proteins and dairy, is essential for the system to be able to provide actual meal planning. This requires a diverse dataset and potentially an approach where the model can detect multiple classes in one image.

G.2 Offline capabilities

As discussed in section 14.4, offline capabilities requirements from section 6.2.2 are not met. The implementation of these offline requirements causes the system to live up to two additional requirements, thus creating a better experience for the user when experiencing internet connection issues. This implementation should be a priority of the future work as it is one of reason for choosing the Firestore database from Firebase JS SDK, as discussed in section 8.3. By implementing offline capabilities, the user would be able to change their preferences when internet connection is lost, with the preferences being sent to the database when internet connection is reestablished.